Comments: Some students gave the force instead of the field strength as the answer.

4 (a) (i) $T = 2\pi\sqrt{(2.3 / 63)} = 1.20$ s — C1

$\omega = 2\pi / T = 2\pi / 1.20 = 5.23$ rad s^{-1} — A1

OR directly from $\omega = \sqrt{(k / m)}$

(ii) correct substitution — C1

giving $E = ½ \times 2.3 \times 5.23^2 \times 0.28^2 = 2.47$ J — A1

Comments: This part was answered well.

(b)

	kinetic energy / J	gravitational potential energy / J	elastic potential energy / J	total energy / J
top	**0**	**6.32**	–3.85	**2.47**
middle	**2.47**	reference zero	reference zero	**2.47**
bottom	**0**	**– 6.32**	**8.79**	**2.47**

kinetic energy column correct — B1

$mgh = 2.3 \times 9.81 \times 0.28 = 6.32$ J — B1

giving + 6.32 at top and – 6.32 at bottom — B1

total energy constant at 6.32 – 3.85 = 2.47 J — B1

so e.p.e. at bottom = 8.79 J — B1

Comments: No error-carried-forward for this part of the question. Total energy can be calculated using the first row after obtaining 6.32 J for GPE.
Some students gave the EPE at the bottom to be +3.85 J.

(c) (i) at the resonant frequency $\omega = 2\pi f = 2\pi \times 35.5 = 223$ rad s^{-1} — C1

use of $A = 0.0114$ m in equation $a = \omega^2 A$

$= (223)^2 (0.0114) = 567$ m s^{-2} — A1

Comments: Students should use f = 35.5 Hz as given in the question and not to read from graph.

(ii) same starting point and lower graph peak — B1

maximum amplitude (at same or lower frequency)

within original shape	B1

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